INTERNAL · CONFIDENTIAL DRAFT

Chat With Your Supply Chain Data

Technical Design Document

High-level and low-level design for a conversational interface over the legacy supply-chain database.

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Status Draft for review

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1 Purpose & Business Context

The supply chain team's historical operational data lives in an older, protected database system with no public interface other applications can plug into. Today, answering an open-ended question about that data means one of two things: using a fixed dashboard, which can only answer the questions it was originally built for, or asking someone who can write a database query by hand.

This document describes the design for a conversational tool that lets people ask questions about that data in plain English and receive an answer grounded in the real records — without ever giving that tool the ability to change or delete anything.

2 Scope

In scope

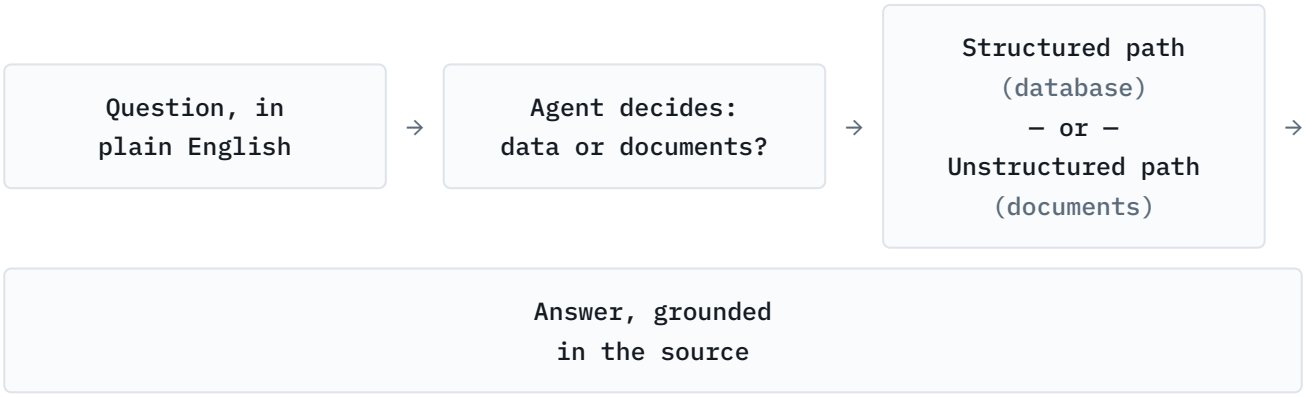
Read access to the logistics data (SQL Server) and to standard operating procedure documents (SOPs). A conversational agent that answers questions using both. Development and testing in a local environment before any connection to a live client system.

Out of scope, for now

Writing or modifying any data. Connecting to the client's production system. Content-moderation style guardrails — the reasoning is covered in Section 6.

3 High-Level Design

Two different filing systems need two different ways of searching them. A spreadsheet with tens of thousands of rows is searched with filters and lookups; a shelf of procedure documents is searched by reading the relevant pages. The agent's job is to recognize which kind of search a question needs, and to do it automatically.



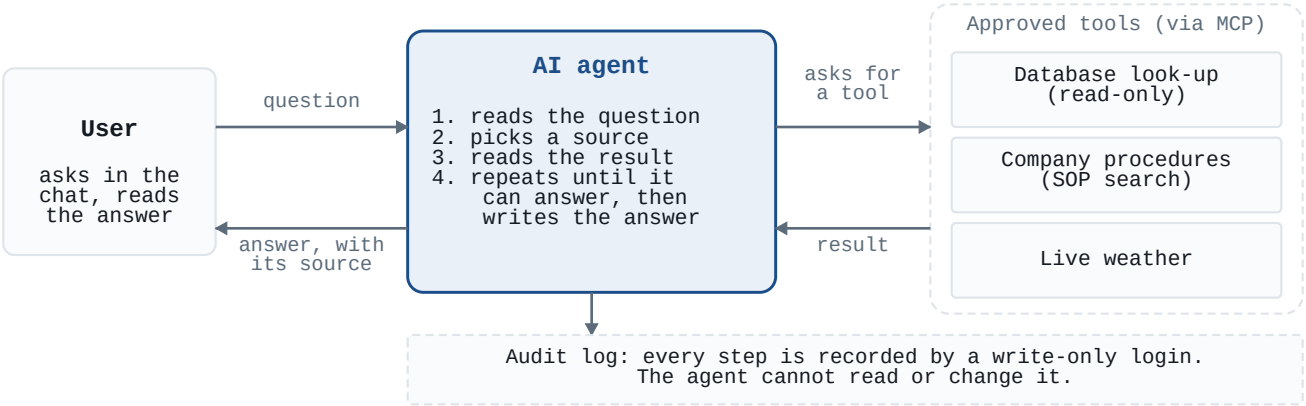
Both paths share one rule: the agent's access to the data is read-only, regardless of which path is used. Section 5 covers how that is enforced.

Why one search method doesn't cover both

The logistics dataset is too large to hand the agent in full for every question, so it is searched the way a database is normally searched — with a query — rather than being read end to end. The SOPs are a small set of documents, so they are searched by meaning, the way a person would search a shared drive.

How one question is answered, end to end

The agent works in a loop: it asks for one source at a time, reads what comes back, and only answers once it has enough. Every step is also written to an audit log.

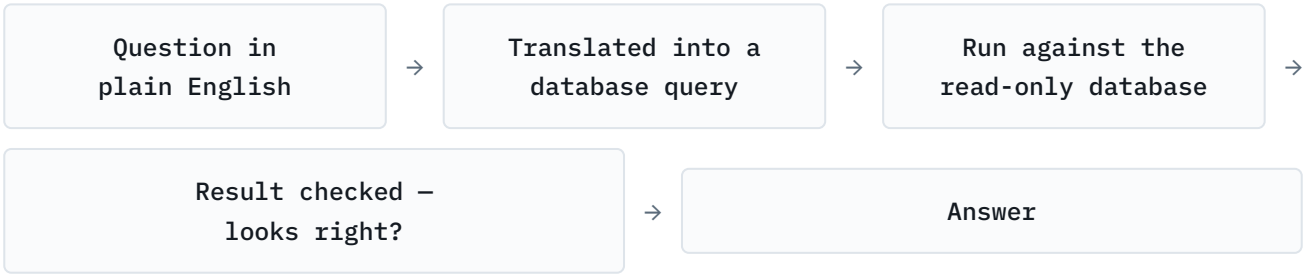


4 Low-Level Design

4.1 The structured data path

> **text2sql**
> In plain terms: a translator that turns a plain-English question into the technical query language the database understands.

> **MCP (Model Context Protocol)**
> In plain terms: a controlled way for the agent to call a small set of approved tools, instead of having open-ended access to the system.



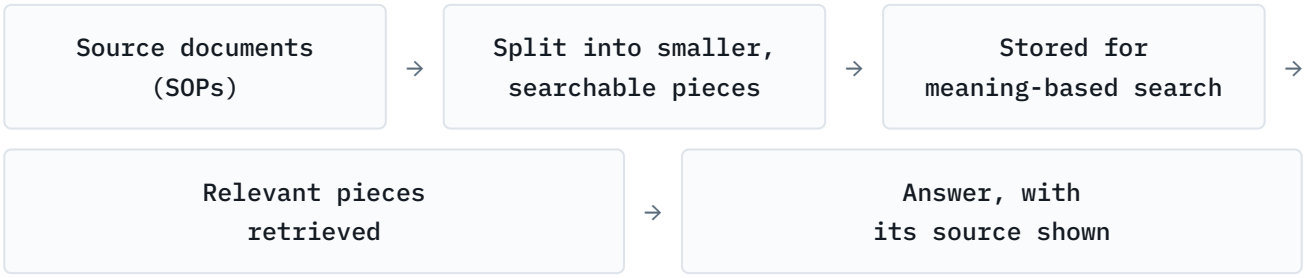
If a result looks off, the agent revises the query and tries again before answering — the loop back into the third step above.

Trade-off. This approach is fast to build and isn't limited to a fixed set of preset views. The cost is that a translated query can occasionally be wrong, which is why the agent is designed to check its own results and retry, rather than trusting the first attempt blindly.

Looking ahead. As the underlying dataset grows, the storage and indexing behind this path will need to scale with it. That is a forward planning item, not a current limitation.

4.2 The unstructured data path

> **RAG (Retrieval-Augmented Generation)**
> In plain terms: search first, then write the answer using only what was found — and always show where it came from.



Three items in this path are open and not yet finalized:

- How documents should be split into pieces (the "chunking strategy").
- What happens when a source document is edited or removed after it has already been indexed.
- The pipeline for bringing in new documents (PDFs, markdown files) still needs to be built.

4.3 What people can ask

Location or coordinates · perishable goods and temperature · weather conditions · historical demand · supply chain disruptions · any combination of the above · and direct questions about the SOPs — all in the same conversation.

5 Governance & Security

The agent is designed so that even if something went wrong upstream — a mistaken instruction, an attempted misuse — it still could not change or delete data. That protection lives in the database itself, not in the application's own code.

Least-privilege access

The login the agent uses is restricted to look-up (read) operations only.

Governed views

The agent is only shown pre-approved, relevant slices of the data, not the raw underlying tables.

Immutable audit log

Every question asked and every answer given is recorded permanently, for later review.

Role clarity

The agent never gives someone access to more information than that person is already allowed to see.

6 Guardrails — a Deliberate Scope Decision

Content-moderation style guardrails are not being added at this stage. That is a deliberate choice, not an oversight: guardrails matter most for tools used by the public, or where customer-identifying data is involved. This is an internal tool used by internal staff, so that risk is lower today. The decision would be revisited if the audience or the sensitivity of the data changes.